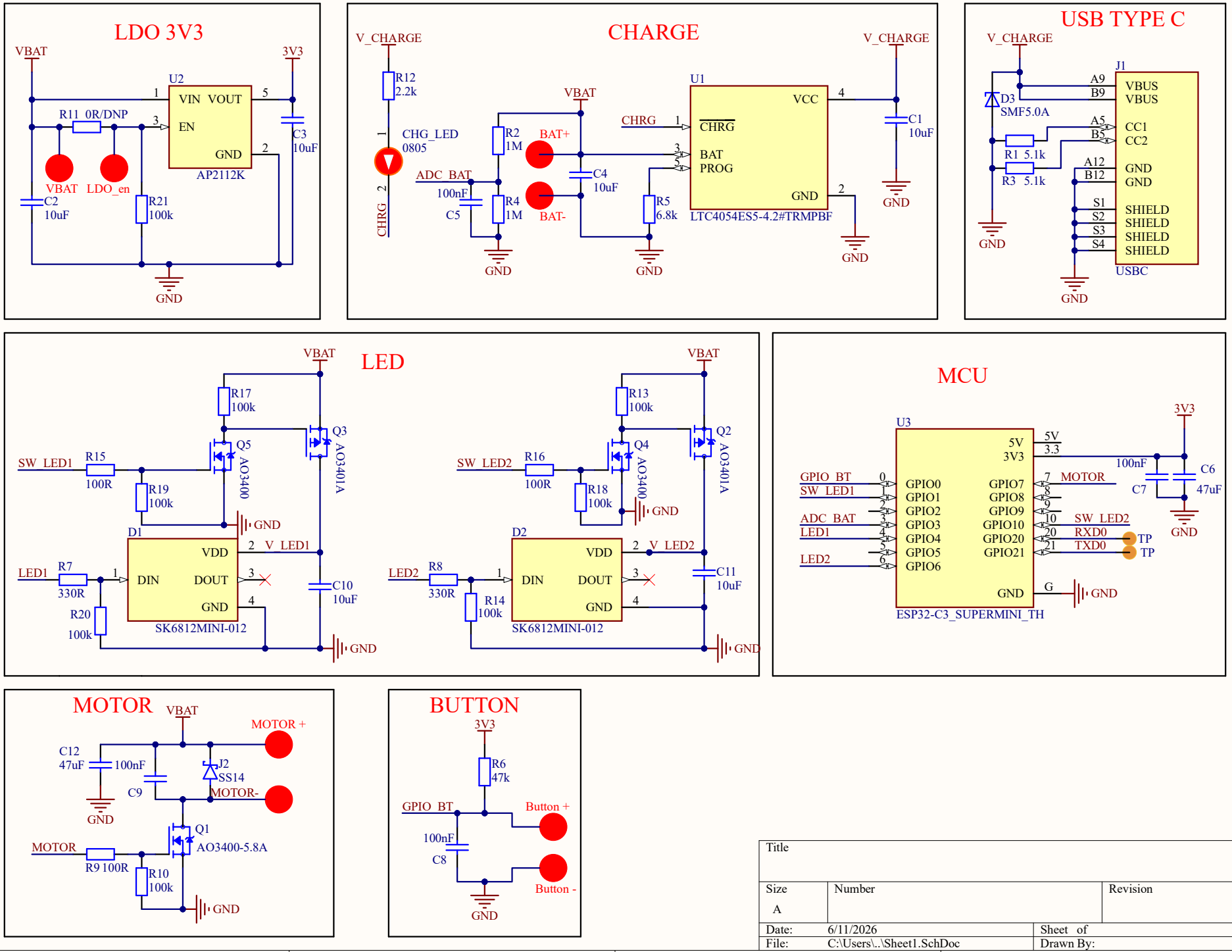

LDO 3V3
VBAT
3V3
U2
VIN VOUT
EN
GND
AP2112K
R11 0R/DNP
C3 10uF
C2 10uF
R21 100k
VBAT LDO_en
GND
CHARGE
V_CHARGE
R12 2.2k
CHG_LED 0805
CHRG
ADC_BAT
100nF C5
R2 1M
R4 1M
BAT+
BAT-
C4 10uF
R5 6.8k
U1
CHRG
BAT
PROG
VCC
GND
LTC4054ES5-4.2#TRMPBF
C1 10uF
USB TYPE C
J1
A9 B9 VBUS
A5 CC1
B5 CC2
A12 B12 GND
S1 S2 S3 S4 SHIELD
USBC
D3 SMF5.0A
R1 5.1k
R3 5.1k
LED
SW_LED1
SW_LED2
R15 100R
R16 100R
R17 100k
R18 100k
R19 100k
R13 100k
Q5 AO3400
Q4 AO3400
Q3 AO3401A
Q2 AO3401A
D1
D2
VDD
DIN
DOUT
GND
V_LED1
V_LED2
SK6812MINI-012
LED1
LED2
R7 330R
R8 330R
R20 100k
R14 100k
C10 10uF
C11 10uF
MCU
U3
GPIO0 GPIO1 GPIO2 GPIO3 GPIO4 GPIO5 GPIO6
GPIO7 GPIO8 GPIO9 GPIO10 GPIO20 GPIO21
5V 3V3
3.3
GPIO_BT
ADC_BAT
MOTOR
SW_LED2
RXD0
TXD0
TP
C6 47uF
C7 100nF
ESP32-C3_SUPERMINI_TH
MOTOR
MOTOR +
MOTOR-
C12 47uF
C9 100nF
J2 SS14
Q1 AO3400-5.8A
R9 100R
R10 100k
BUTTON
R6 47k
Button +
Button -
C8 100nF
Title
Size A
Number
Revision
Date: 6/11/2026
Sheet of
File: C:\Users\..\Sheet1.SchDoc
Drawn By: